Your security scans passed. Your deployment was approved. But what happens when an attacker breaks in after production?

What is Runtime Threat Detection?

- Security monitoring during application execution
- Detects suspicious behavior in real time
- Focuses on what is happening right now inside systems and containers
- Last line of defense when preventive controls fail

Why Runtime Security is Important?

- Not all attacks are prevented before deployment
- New vulnerabilities can be exploited at runtime
- Misconfigurations and insider threats can occur anytime
- Helps security teams detect attacks immediately

Traditional Security vs Runtime Security

- **Traditional Security**
- Focuses on prevention
- Before deployment
- SAST, SCA, DAST, SBOM, IaC Scanning
- **Runtime Security**
- Focuses on detection and response
- After deployment
- Monitors live workloads and user activities

What Should Be Monitored at Runtime

- Unauthorized shell access
- Privilege escalation attempts
- Suspicious process execution
- Unexpected network connections
- Sensitive file access
- Container escape attempts

Common Runtime Threats

- Reverse shells
- Malware execution
- Cryptomining attacks
- Unauthorized privilege escalation
- Container breakout attempts
- Lateral movement inside infrastructure

What is Policy-Based Detection?

- Security rules define suspicious behavior
- System continuously compares activities against policies
- Alerts generated when violations occur

What is Falco?

Why Falco is Popular in Industry?

Typical Runtime Security Workflow

Real-World Importance